



Clonucopya

Clonal Evolution & Treatment Report

Study Statistics: CRUKP7127

Overview of study CRUKP7127's key statistics by sample (N = 5). This section summarizes the core characteristics of each sample in the study and serves as the starting point for interpreting the rest of the report. The values reflect each sample's mutational burden, genomic alterations, and the number of therapeutic options identified (total drugs and Best Therapeutic Candidates, BTC). The exact set of metrics shown may vary slightly depending on the analysis settings selected for this run. Refer to the table below for the specific indicators reported.

This report was generated in clinical mode. In clinical mode, results are filtered more stringently, excluding drugs with experimental status and pathway member interaction type. In discovery mode, no filters are applied, and all drug hits identified are displayed regardless of their experimental status or interaction type.

All definitions are provided in the Glossary of Terms.

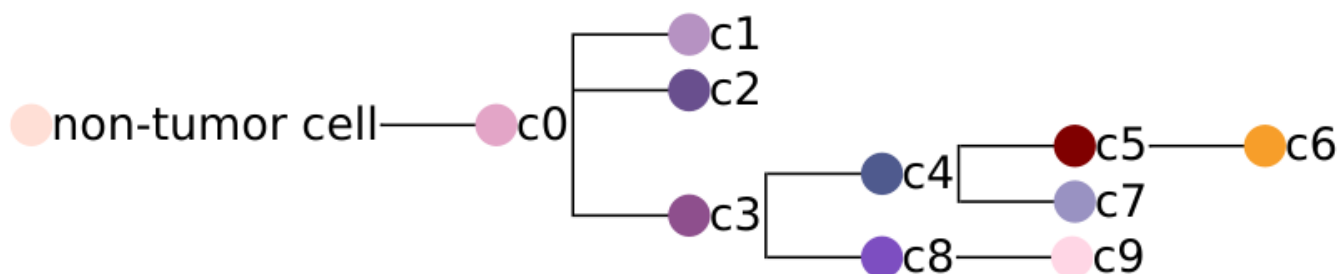
The source files are available at: results/CRUKP7127/report/components/report_panel*.tsv.

Sample Name BR_LN01	Sex female	Tumor Content 0.45	Total Drugs 171
Study CRUKP7127	Number of mutations 134	Number of clones 5	BTCs 0
Sample Name SU_T1-R1	Sex female	Tumor Content 0.61	Total Drugs 256
Study CRUKP7127	Number of mutations 84	Number of clones 5	BTCs 0
Sample Name SU_T1-R2	Sex female	Tumor Content 0.46	Total Drugs 167
Study CRUKP7127	Number of mutations 130	Number of clones 2	BTCs 0
Sample Name SU_T1-R3	Sex female	Tumor Content 0.42	Total Drugs 357
Study CRUKP7127	Number of mutations 346	Number of clones 7	BTCs 35
Sample Name SU_T1-R4	Sex female	Tumor Content 0.24	Total Drugs 319
Study CRUKP7127	Number of mutations 341	Number of clones 7	BTCs 35

Tumor Phylogeny

Based on the variant allele frequency (VAF) and changes in copy number, the evolutionary hierarchy of the tumor has been inferred. The phylogenetic tree illustrates the relationships among the different cellular subpopulations (clones).

The source files are available at: results/CRUKP7127/report/components/clonal_tree.png.

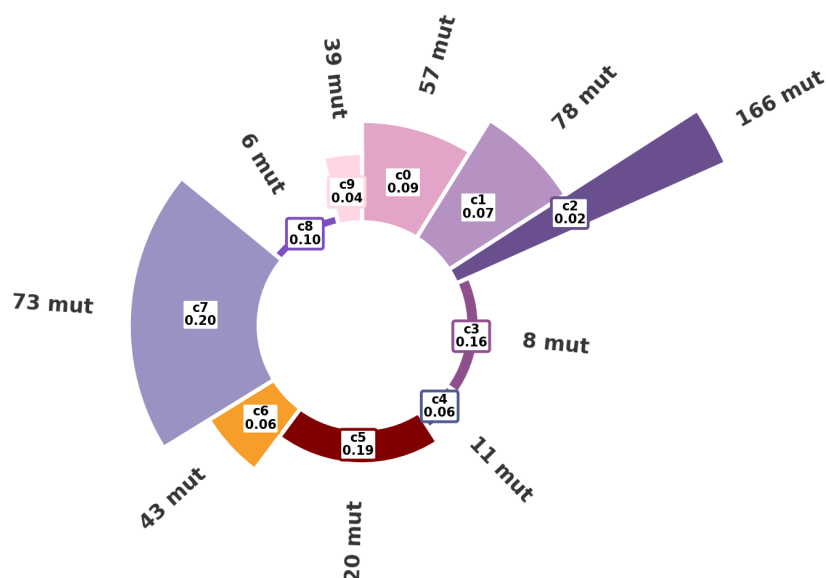


Clonal composition

This is a rose (nightingale) chart showing the clonal and mutational composition at the study level. Each bar represents one clone: its width reflects clonal prevalence (the clone's weight within the tumor's overall clonal structure), and its height reflects the number of mutations that define that clone's identity. Note that prevalence and mutation count are not directly correlated. A clone can carry many mutations yet represent only a small fraction of the tumor, or vice versa, depending on each mutation's biological impact.

The prevalence values across clones may not sum to 100%. This gap represents the undetermined portion of the clonal structure, which can result from normal-cell contamination (already accounted for via the tumor purity estimate), statistical uncertainty in prevalence estimation, or rounding.

The source files are available at: [results/CRUKP7127/report/components/study_name}_clonal_composition.png](#).



Drug Summary

The Drug Summary provides a high-level overview of the therapeutic candidates identified across the entire study. For each drug, the table reports the number of genetic alterations supporting its prioritization, its regulatory approval status (APPROVED, CLINICAL_TRIALS, or EXPERIMENTAL), and the type of interaction with the affected genes (DIRECT TARGET, BIOMARKER, or PATHWAY_MEMBER). This table summarizes up to 50 top-ranked drug candidates (25 sensitive and 25 resistance when available) derived from the mutational landscape of all samples included in the study. A detailed per-clone breakdown is available in the Drug Prioritization section.

Analysis mode: Clinical (standard filtering applied — experimental-status drugs and pathway-member interactions excluded).

The response of each drug is evaluated using the DScore (range -1 to +1):

- Sensitivity (DScore > 0): Positive values indicate that the mutations confer sensitivity to the drug (optimal candidates with a DScore > 0.7 are highlighted in bold).
- Resistance (DScore < 0): Negative values suggest that the clone possesses mutations associated with therapeutic resistance.

Drug-Gene Interactions:

- DT (Direct Target): The drug acts directly on the mutated protein.
- BM (Biomarker): The alteration acts as a biomarker predictive of response.
- PM (Pathway Member): The mutation affects the signaling pathway targeted by the drug.

The table below provides a simplified overview of the drugs targeting the affected clones and the specific genes. Drugs are prioritized using the following criteria: (1) clone coverage, (2) maximum dScore of the drug regarding the targeted genes (dScore), and (3) interaction type importance (DT>BM>PM).

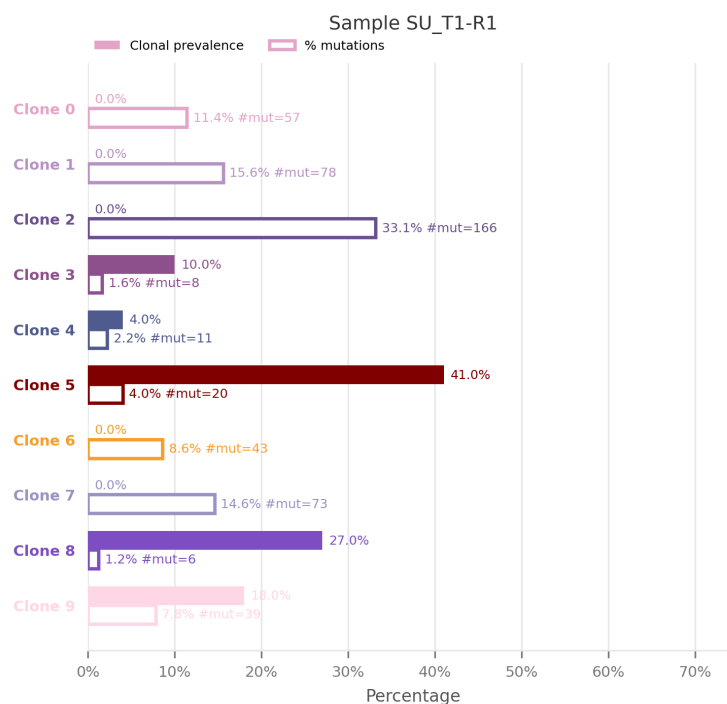
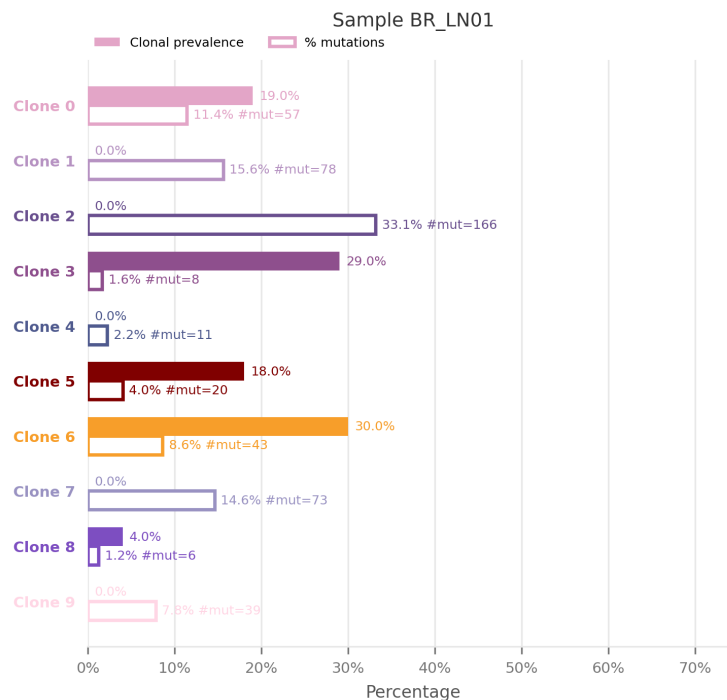
The source files are available at: results/CRUKP7127/report/components/drug_summary.tsv.

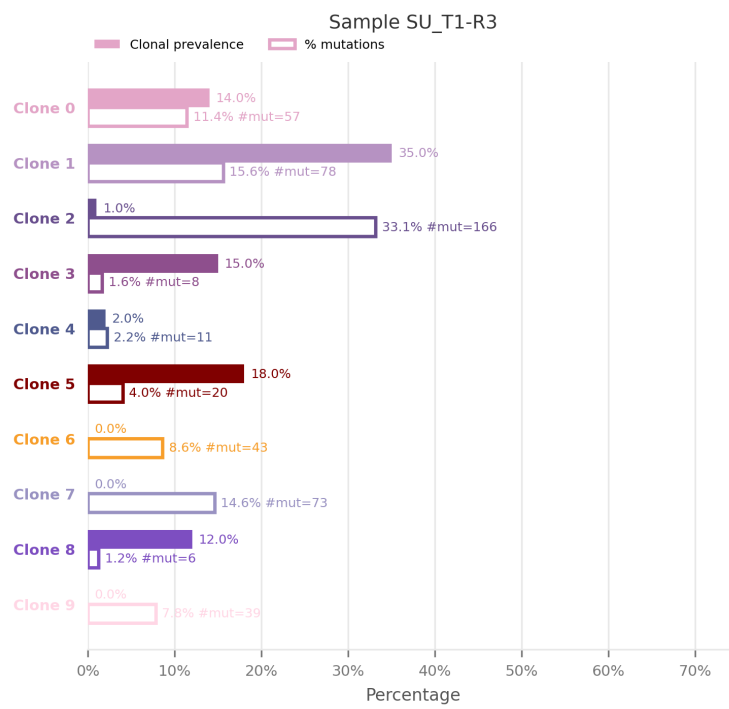
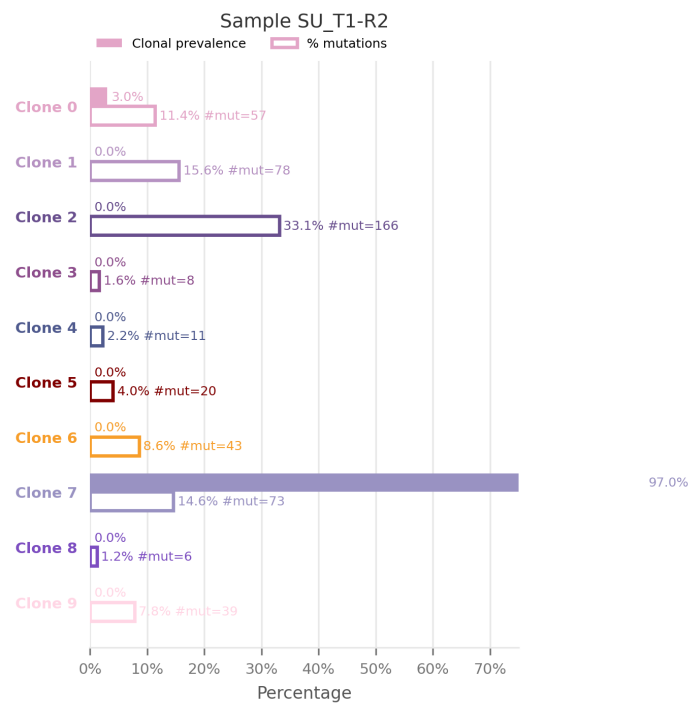
Drug	Status	Max. dScore	Target Clones	Gene Interactions	Drug Response
ALPELISIB	APPROVED	0.920	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
COPANLISIB	APPROVED	0.919	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
ALPELISIB + FULVESTRANT	APPROVED	0.913	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
NVP-BGJ398	APPROVED	0.913	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
IDELALISIB	APPROVED	0.916	Total: 5 BM: 2; PM: 5, 6, 7, 9	BM: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
DUVELISIB	APPROVED	0.914	Total: 5 BM: 2; PM: 5, 6, 7, 9	BM: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
DACTOLISIB	CLINICAL_TRIALS	0.528	Total: 5 BM: 7; DT: 2; PM: 5, 6, 7, 9	BM: COL5A1; DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
957054-30-7	CLINICAL_TRIALS	0.527	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
GEDATOLISIB	CLINICAL_TRIALS	0.517	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
PILARALISIB	CLINICAL_TRIALS	0.517	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
BUPARLISIB	CLINICAL_TRIALS	0.516	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
PF-04691502	CLINICAL_TRIALS	0.516	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
SONOLISIB	CLINICAL_TRIALS	0.516	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
APITOLISIB	CLINICAL_TRIALS	0.515	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
SERABELISIB	CLINICAL_TRIALS	0.515	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
SAMOTOLISIB	CLINICAL_TRIALS	0.514	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
XL-765	CLINICAL_TRIALS	0.514	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
CH5132799	CLINICAL_TRIALS	0.513	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
GSK2636771	CLINICAL_TRIALS	0.513	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
TASELISIB	CLINICAL_TRIALS	0.125	Total: 5 DT: 2; PM: 5, 6, 7, 9	DT: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
GENISTEIN	CLINICAL_TRIALS	0.522	Total: 5 BM: 0, 1; PM: 2, 6, 9	BM: TJP1; PM: MAPK3, NRG1, PIK3CA	SENSITIVITY
AZD8186	CLINICAL_TRIALS	0.513	Total: 5 BM: 2; PM: 5, 6, 7, 9	BM: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
ETHANOL	CLINICAL_TRIALS	0.512	Total: 5 BM: 0, 1, 5; DT: 2, 7	BM: GCKR, THSD7B, TJP1; DT: CACNA1C, GABRG1	SENSITIVITY
OMIPALISIB	CLINICAL_TRIALS	0.115	Total: 5 BM: 2; PM: 5, 6, 7, 9	BM: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
ACALISIB	CLINICAL_TRIALS	0.113	Total: 5 BM: 2; PM: 5, 6, 7, 9	BM: PIK3CA; PM: LAMB4, MAPK3, NRG1, STAT2	SENSITIVITY
DACTINOMYCIN	APPROVED	-0.811	Total: 1 BM: 7	BM: COL5A1	RESISTANCE

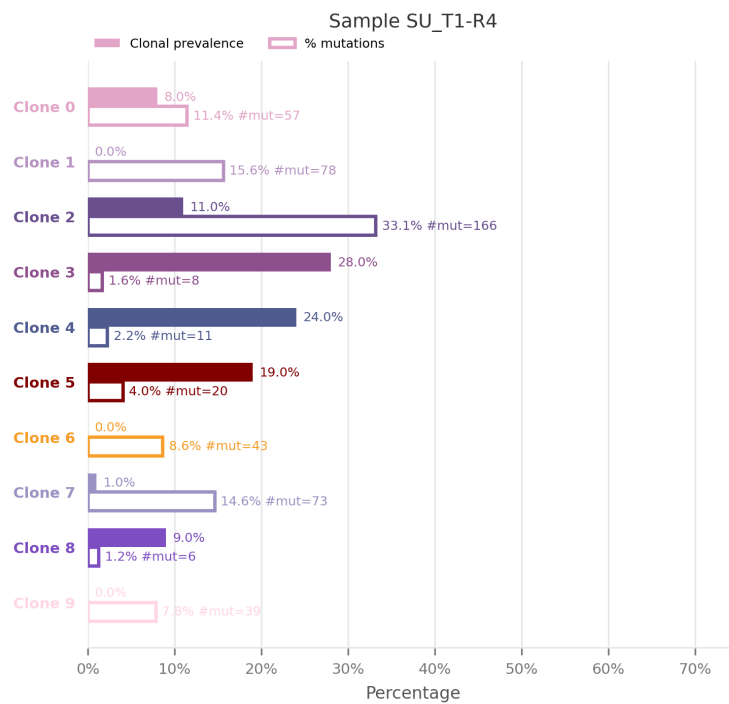
Sample-Level Clonal Structure Analysis

Analysis of clonal evolution patterns and proportions across samples.

The source files are available at: [results/CRUKP7127/report/components/mutation_contribution/{sample_id}_mutation_contribution.png](https://results.CRUKEP7127/report/components/mutation_contribution/{sample_id}_mutation_contribution.png).



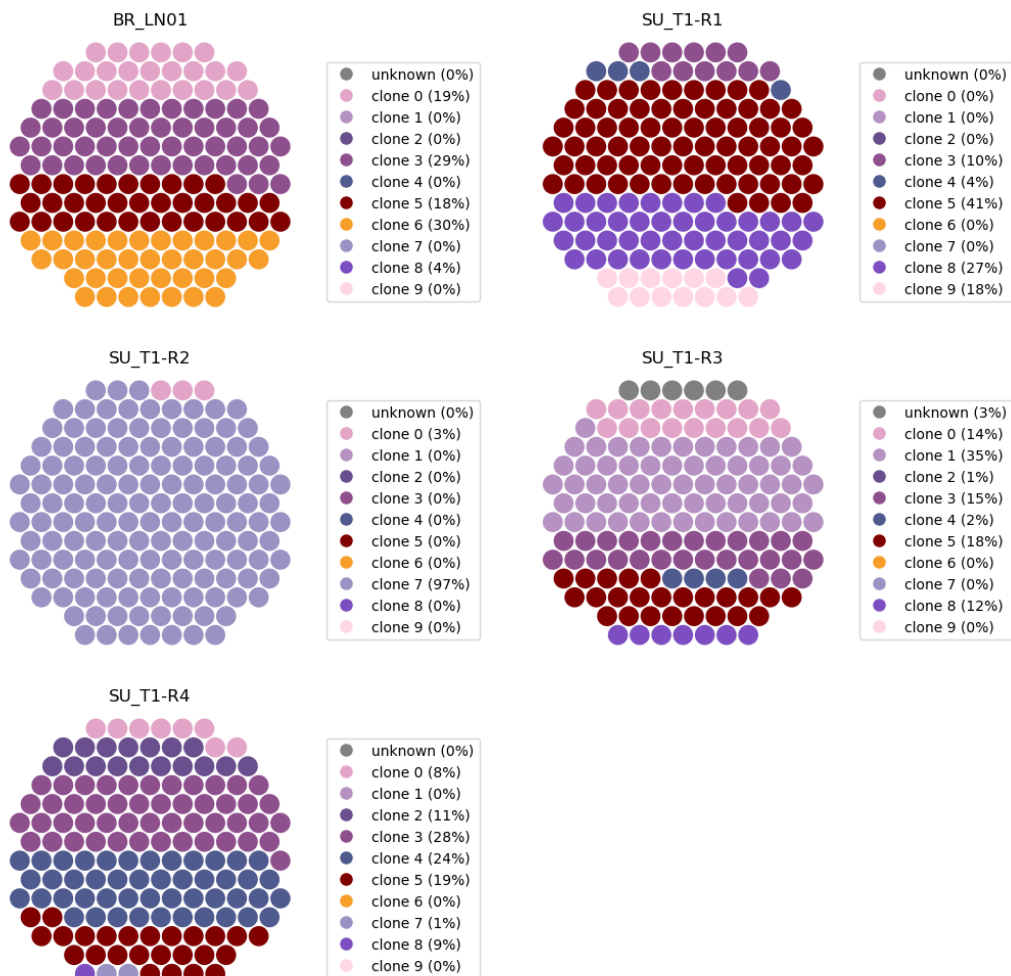




Clonal Proportions

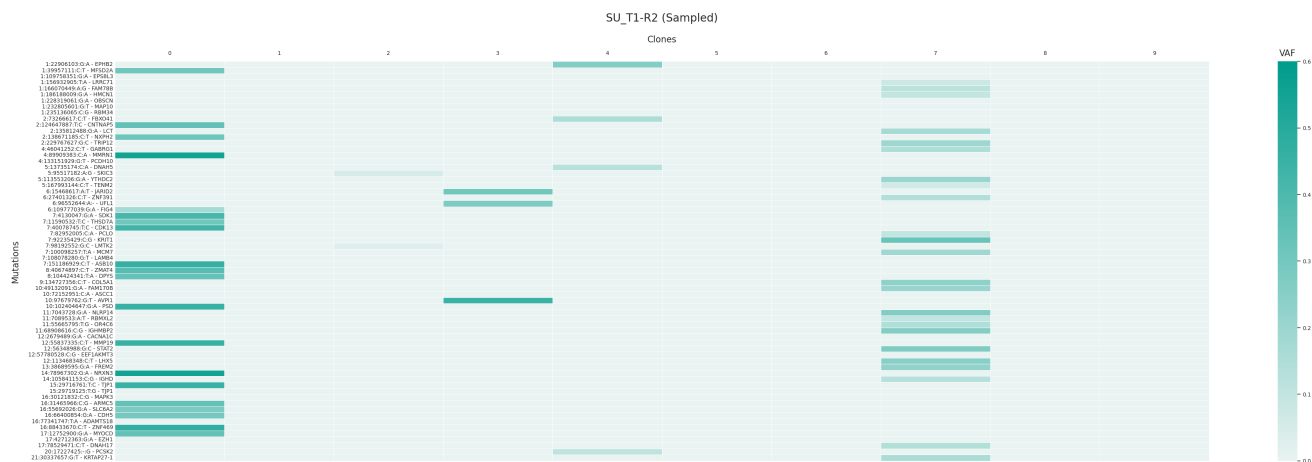
Based on the inferred tumor phylogeny, the spheres representations show the distribution and percentage of each clone across the different biopsies or time points, allowing us to visualize which clones dominate the tumor tissue.

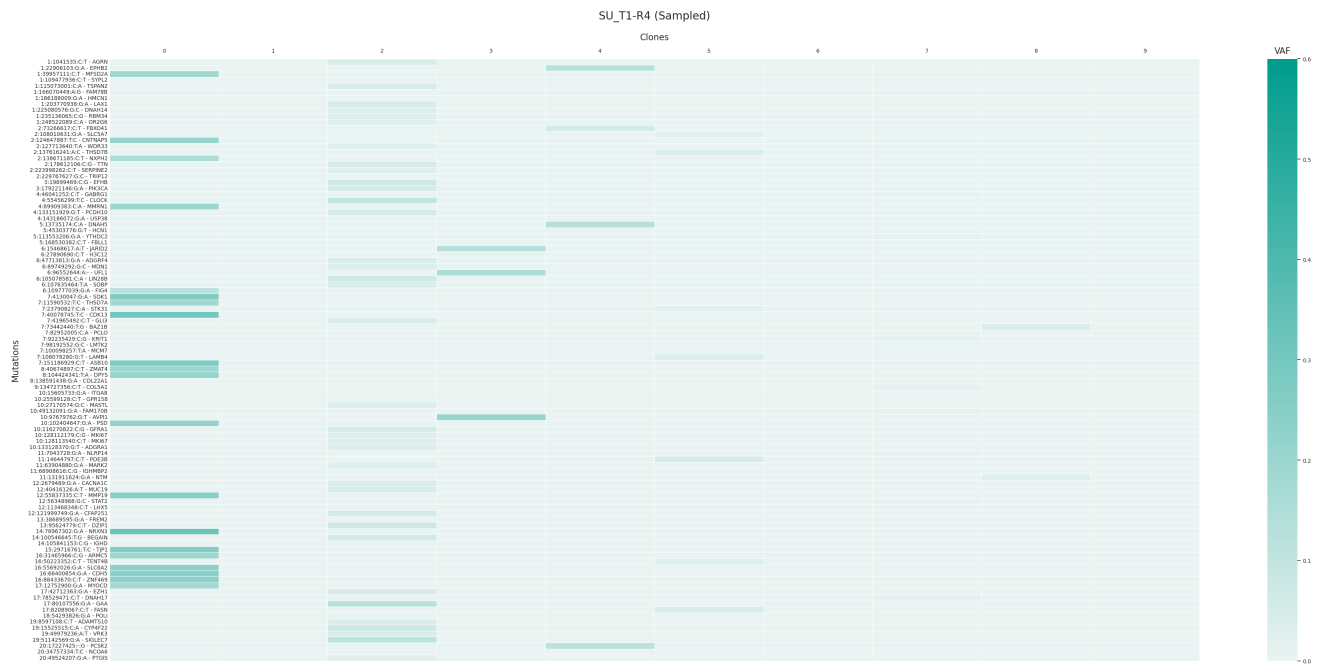
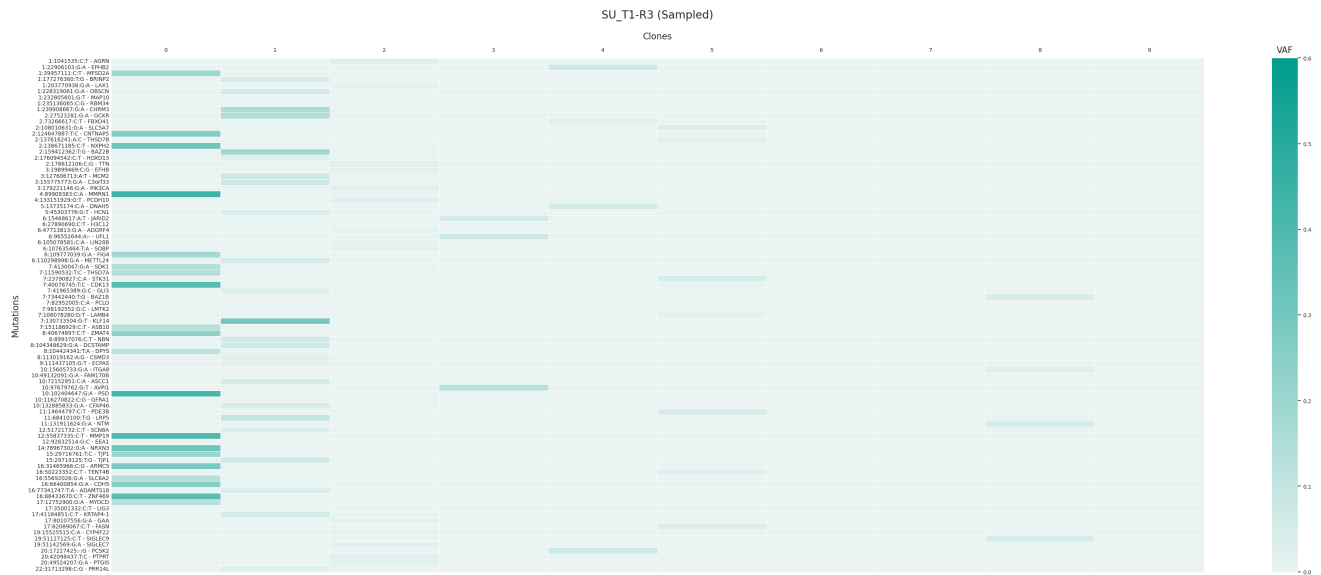
The source files are available at: results/CRUKP7127/report/components/spheres_of_clones/{sample_id}_sphere_of_clones.png.



Clonal Alterations

The source files are available at: `results/CRUKP7127/report/components/vaf_heatmaps/sampled/{sample_id}_sampled_vaf_heatmap.png`.





Pharmacogenomics

Gene Alterations

Both SNVs and small indels (if selected at the beginning of the workflow) used in clone inference are displayed with the most relevant information at the clone and sample level. To sum up the results, this section only shows top 10 mutations per clone with moderate or high impact.

Mutations are sorted using the following criteria:

- Clone.
- Mutation positon.
- Impact: High and Moderate.

The source files are available at: [results/CRUKP7127/report/components/gene_alterations.tsv](#).

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
0	1:39957111:C:T	MFSD2A	ENSG00000168389	BR_LN01: 0.366; SU_T1-R1: 0.424; SU_T1-R2: 0.306; SU_T1-R3: 0.195; SU_T1-R4: 0.191	stop_gained	HIGH

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
0	2:124647887:T:C	CNTNAP5	ENSG00000155052	BR_LN01: 0.297; SU_T1-R1: 0.419; SU_T1-R2: 0.342; SU_T1-R3: 0.260; SU_T1-R4: 0.214	missense_variant	MODERATE
0	2:138671185:C:T	NXPH2	ENSG00000144227	BR_LN01: 0.284; SU_T1-R1: 0.369; SU_T1-R2: 0.317; SU_T1-R3: 0.318; SU_T1-R4: 0.161	missense_variant	MODERATE
0	4:89909383:C:A	MMRN1	ENSG00000138722	BR_LN01: 0.468; SU_T1-R1: 0.692; SU_T1-R2: 0.549; SU_T1-R3: 0.434; SU_T1-R4: 0.197	missense_variant	MODERATE
0	6:109777039:G:A	FIG4	ENSG00000112367	BR_LN01: 0.266; SU_T1-R1: 0.424; SU_T1-R2: 0.171; SU_T1-R3: 0.183; SU_T1-R4: 0.113	missense_variant	MODERATE
0	7:4130047:G:A	SDK1	ENSG00000146555	BR_LN01: 0.350; SU_T1-R1: 0.389; SU_T1-R2: 0.411; SU_T1-R3: 0.136; SU_T1-R4: 0.265	missense_variant	MODERATE
0	7:11590532:T:C	THSD7A	ENSG00000005108	BR_LN01: 0.363; SU_T1-R1: 0.481; SU_T1-R2: 0.307; SU_T1-R3: 0.141; SU_T1-R4: 0.192	missense_variant	MODERATE
0	7:40078745:T:C	CDK13	ENSG00000065883	BR_LN01: 0.499; SU_T1-R1: 0.494; SU_T1-R2: 0.436; SU_T1-R3: 0.375; SU_T1-R4: 0.304	missense_variant	MODERATE
0	7:151186929:C:T	ASB10	ENSG00000146926	BR_LN01: 0.270; SU_T1-R1: 0.355; SU_T1-R2: 0.458; SU_T1-R3: 0.126; SU_T1-R4: 0.276	missense_variant	MODERATE
0	8:40674897:C:T	ZMAT4	ENSG00000165061	BR_LN01: 0.191; SU_T1-R1: 0.393; SU_T1-R2: 0.376; SU_T1-R3: 0.229; SU_T1-R4: 0.200	missense_variant	MODERATE
1	8:89937076:C:T	NBN	ENSG00000104320	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.059; SU_T1-R4: 0.000	splice_acceptor_variant	HIGH
1	8:104348629:G:A	DCSTAMP	ENSG00000164935	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.068; SU_T1-R4: 0.000	stop_gained	HIGH
1	9:111437105:G:T	ECPAS	ENSG00000136813	BR_LN01: 0.000; SU_T1-R1: 0.003; SU_T1-R2: 0.000; SU_T1-R3: 0.027; SU_T1-R4: 0.000	stop_gained	HIGH
1	1:177276360:T:G	BRINP2	ENSG00000198797	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.047; SU_T1-R4: 0.000	missense_variant	MODERATE
1	1:228319061:G:A	OBSCN	ENSG00000154358	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.002; SU_T1-R3: 0.059; SU_T1-R4: 0.000	missense_variant	MODERATE
1	1:239908667:G:A	CHRM3	ENSG00000133019	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.156; SU_T1-R4: 0.000	missense_variant	MODERATE
1	2:27523281:G:A	GCKR	ENSG00000084734	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.154; SU_T1-R4: 0.000	missense_variant	MODERATE

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
1	2:159412362:T:G	BAZ2B	ENSG00000123636	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.191; SU_T1-R4: 0.000	missense_variant	MODERATE
1	2:176094542:C:T	HOXD13	ENSG00000128714	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.030; SU_T1-R4: 0.000	missense_variant	MODERATE
1	3:127606713:A:T	MCM2	ENSG00000073111	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.074; SU_T1-R4: 0.000	missense_variant	MODERATE
2	1:115073001:C:A	TSPAN2	ENSG00000134198	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.045	stop_gained	HIGH
2	1:203770938:G:A	LAX1	ENSG00000122188	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.012; SU_T1-R4: 0.045	splice_donor_variant,non_coding_transcript_variant	HIGH
2	6:107635464:T:A	SOBP	ENSG00000112320	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.014; SU_T1-R4: 0.041	stop_lost	HIGH
2	17:35001332:C:T	LIG3	ENSG00000005156	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.002; SU_T1-R4: 0.000	stop_gained	HIGH
2	17:42712363:G:A	EZH1	ENSG00000108799	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.004; SU_T1-R3: 0.000; SU_T1-R4: 0.049	stop_gained	HIGH
2	17:80107556:G:A	GAA	ENSG00000171298	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.013; SU_T1-R4: 0.128	splice_acceptor_variant	HIGH
2	19:35723872:C:T	KMT2B	ENSG00000272333	BR_LN01: 0.000; SU_T1-R1: 0.001; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	stop_gained	HIGH
2	1:1041535:C:T	AGRN	ENSG00000188157	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.027; SU_T1-R4: 0.051	missense_variant	MODERATE
2	1:109758351:G:A	EPS8L3	ENSG00000198758	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.002; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
2	1:225080576:G:C	DNAH14	ENSG00000185842	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.031	missense_variant	MODERATE
3	6:96552644:A:-	UFL1	ENSG00000014123	BR_LN01: 0.228; SU_T1-R1: 0.280; SU_T1-R2: 0.279; SU_T1-R3: 0.073; SU_T1-R4: 0.162	frameshift_variant	HIGH
3	6:15468617:A:T	JARID2	ENSG00000008083	BR_LN01: 0.250; SU_T1-R1: 0.206; SU_T1-R2: 0.306; SU_T1-R3: 0.059; SU_T1-R4: 0.137	missense_variant	MODERATE
3	10:97679762:G:T	AVPI1	ENSG00000119986	BR_LN01: 0.430; SU_T1-R1: 0.592; SU_T1-R2: 0.448; SU_T1-R3: 0.124; SU_T1-R4: 0.209	missense_variant	MODERATE

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
4	20:17227425:-:G	PCSK2	ENSG00000125851	BR_LN01: 0.226; SU_T1-R1: 0.204; SU_T1-R2: 0.110; SU_T1-R3: 0.078; SU_T1-R4: 0.113	frameshift_variant	HIGH
4	1:22906103:G:A	EPHB2	ENSG00000133216	BR_LN01: 0.171; SU_T1-R1: 0.101; SU_T1-R2: 0.262; SU_T1-R3: 0.074; SU_T1-R4: 0.128	missense_variant	MODERATE
4	2:73266617:C:T	FBXO41	ENSG00000163013	BR_LN01: 0.110; SU_T1-R1: 0.201; SU_T1-R2: 0.153; SU_T1-R3: 0.019; SU_T1-R4: 0.056	missense_variant	MODERATE
4	5:13735174:C:A	DNAH5	ENSG00000039139	BR_LN01: 0.245; SU_T1-R1: 0.208; SU_T1-R2: 0.123; SU_T1-R3: 0.057; SU_T1-R4: 0.130	missense_variant	MODERATE
5	2:108010631:G:A	SLC5A7	ENSG00000115665	BR_LN01: 0.173; SU_T1-R1: 0.149; SU_T1-R2: 0.000; SU_T1-R3: 0.034; SU_T1-R4: 0.022	missense_variant	MODERATE
5	2:137616241:A:C	THSD7B	ENSG00000144229	BR_LN01: 0.142; SU_T1-R1: 0.180; SU_T1-R2: 0.000; SU_T1-R3: 0.027; SU_T1-R4: 0.041	missense_variant	MODERATE
5	7:23790827:C:A	STK31	ENSG00000196335	BR_LN01: 0.168; SU_T1-R1: 0.214; SU_T1-R2: 0.000; SU_T1-R3: 0.051; SU_T1-R4: 0.020	missense_variant	MODERATE
5	7:108078280:G:T	LAMB4	ENSG00000091128	BR_LN01: 0.221; SU_T1-R1: 0.187; SU_T1-R2: 0.002; SU_T1-R3: 0.014; SU_T1-R4: 0.035	missense_variant	MODERATE
5	11:14644797:C:T	PDE3B	ENSG00000152270	BR_LN01: 0.208; SU_T1-R1: 0.189; SU_T1-R2: 0.000; SU_T1-R3: 0.049; SU_T1-R4: 0.048	missense_variant	MODERATE
5	16:50223352:C:T	TENT4B	ENSG00000121274	BR_LN01: 0.134; SU_T1-R1: 0.170; SU_T1-R2: 0.000; SU_T1-R3: 0.025; SU_T1-R4: 0.037	missense_variant,NMD_transcript_variant	MODERATE
5	17:82089067:C:T	FASN	ENSG00000169710	BR_LN01: 0.124; SU_T1-R1: 0.124; SU_T1-R2: 0.000; SU_T1-R3: 0.036; SU_T1-R4: 0.049	missense_variant	MODERATE
6	1:2510749:C:T	PANK4	ENSG00000157881	BR_LN01: 0.137; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	1:232805601:G:T	MAP10	ENSG00000212916	BR_LN01: 0.167; SU_T1-R1: 0.000; SU_T1-R2: 0.002; SU_T1-R3: 0.002; SU_T1-R4: 0.000	missense_variant	MODERATE
6	4:154744545:G:C	LRAT	ENSG00000121207	BR_LN01: 0.027; SU_T1-R1: 0.001; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	6:36684352:G:A	CDKN1A	ENSG00000124762	BR_LN01: 0.034; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	7:45648745:G:A	ADCY1	ENSG00000164742	BR_LN01: 0.025; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
6	8:32760291:A:C	NRG1	ENSG00000157168	BR_LN01: 0.047; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	8:138591438:G:A	COL22A1	ENSG00000169436	BR_LN01: 0.127; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.002	missense_variant	MODERATE
6	11:56612948:G:C	OR5M1	ENSG00000255012	BR_LN01: 0.043; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	13:75362226:C:A	TBC1D4	ENSG00000136111	BR_LN01: 0.068; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
6	13:113117542:G:A	F7	ENSG00000057593	BR_LN01: 0.027; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
7	2:229767627:G:C	TRIP12	ENSG00000153827	BR_LN01: 0.002; SU_T1-R1: 0.000; SU_T1-R2: 0.193; SU_T1-R3: 0.000; SU_T1-R4: 0.002	stop_gained	HIGH
7	17:78529471:C:T	DNAH17	ENSG00000187775	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.140; SU_T1-R3: 0.000; SU_T1-R4: 0.007	splice_donor_variant	HIGH
7	1:156932905:T:A	LRRC71	ENSG00000160838	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.079; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
7	1:166070449:A:G	FAM78B	ENSG00000188859	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.118; SU_T1-R3: 0.000; SU_T1-R4: 0.002	missense_variant	MODERATE
7	1:186188009:G:A	HMCN1	ENSG00000143341	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.099; SU_T1-R3: 0.000; SU_T1-R4: 0.002	missense_variant,splice_region_variant	MODERATE
7	2:135812488:G:A	LCT	ENSG00000115850	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.172; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
7	4:46041252:C:T	GABRG1	ENSG00000163285	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.149; SU_T1-R3: 0.000; SU_T1-R4: 0.003	missense_variant,splice_region_variant	MODERATE
7	5:113553206:G:A	YTHDC2	ENSG00000047188	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.204; SU_T1-R3: 0.000; SU_T1-R4: 0.007	missense_variant	MODERATE
7	5:167993144:C:T	TENM2	ENSG00000145934	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.059; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
7	6:27401326:C:T	ZNF391	ENSG00000124613	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.139; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
8	7:73442440:T:G	BAZ1B	ENSG00000009954	BR_LN01: 0.000; SU_T1-R1: 0.095; SU_T1-R2: 0.000; SU_T1-R3: 0.047; SU_T1-R4: 0.034	missense_variant	MODERATE

Clone	Mutation ID	Gene Symbol	Ensembl ID	VAF	Consequence	Impact
8	10:15605733:G:A	ITGA8	ENSG00000077943	BR_LN01: 0.019; SU_T1-R1: 0.228; SU_T1-R2: 0.000; SU_T1-R3: 0.024; SU_T1-R4: 0.002	missense_variant	MODERATE
8	11:131911624:G:A	NTM	ENSG00000182667	BR_LN01: 0.000; SU_T1-R1: 0.249; SU_T1-R2: 0.000; SU_T1-R3: 0.052; SU_T1-R4: 0.029	missense_variant	MODERATE
8	19:51127125:C:T	SIGLEC9	ENSG00000129450	BR_LN01: 0.057; SU_T1-R1: 0.200; SU_T1-R2: 0.000; SU_T1-R3: 0.049; SU_T1-R4: 0.000	missense_variant	MODERATE
9	4:76755194:T:C	SHROOM3	ENSG00000138771	BR_LN01: 0.000; SU_T1-R1: 0.153; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	splice_donor_variant	HIGH
9	7:99929382:G:-	GJC3	ENSG00000176402	BR_LN01: 0.000; SU_T1-R1: 0.114; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	frameshift_variant	HIGH
9	10:25599128:C:T	GPR158	ENSG00000151025	BR_LN01: 0.000; SU_T1-R1: 0.077; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.001	stop_gained	HIGH
9	1:109477936:C:T	SYPL2	ENSG00000143028	BR_LN01: 0.000; SU_T1-R1: 0.040; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.002	missense_variant	MODERATE
9	4:155711048:G:C	GUCY1A1	ENSG00000164116	BR_LN01: 0.000; SU_T1-R1: 0.027; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
9	5:177238432:T:G	NSD1	ENSG00000165671	BR_LN01: 0.000; SU_T1-R1: 0.091; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
9	6:27890690:C:T	H3C12	ENSG00000197153	BR_LN01: 0.000; SU_T1-R1: 0.095; SU_T1-R2: 0.000; SU_T1-R3: 0.001; SU_T1-R4: 0.001	missense_variant	MODERATE
9	7:100892698:C:T	ACHE	ENSG00000087085	BR_LN01: 0.000; SU_T1-R1: 0.131; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE
9	12:92832514:G:C	EEA1	ENSG00000102189	BR_LN01: 0.000; SU_T1-R1: 0.036; SU_T1-R2: 0.000; SU_T1-R3: 0.002; SU_T1-R4: 0.000	missense_variant,splice_region_variant	MODERATE
9	16:30121832:C:G	MAPK3	ENSG00000102882	BR_LN01: 0.000; SU_T1-R1: 0.022; SU_T1-R2: 0.002; SU_T1-R3: 0.000; SU_T1-R4: 0.000	missense_variant	MODERATE

Drug Prioritization

The small variant analysis performed by PanDrugs2 displays only mutations classified as clinically relevant.

The vulnerability of each tumor clone to different drugs is assessed by considering only mutations with a relevant functional impact. Therapeutic options are prioritized by evaluating two fundamental dimensions: GScore (Biological Target) and DScore (Clinical Evidence). Check Drug Summary section to see how the scoring works.

We consider drugs with high clinical potential (DScore ≥ 0.7) that target key driver genes (GScore ≥ 0.6) to be Top Therapeutic Candidates (BTC), highlighted in bold.

The table below provides a simplified overview of the drugs targeting the affected genes, with the top 3 drugs selected per genetic alteration and ranked by Status and dScore. Drugs are sorted using the following criteria: (1) clone, (2) mutation position, (3) status, and (4) interaction type.

All definitions are provided in the Glossary of Terms.

The source files are available at: results/CRUKP7127/report/components/gene_alterations.tsv.

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
0	12:55837335:C:T	MMP19	BR_LN01: 0.496; SU_T1-R1: 0.576; SU_T1-R2: 0.456; SU_T1-R3: 0.391; SU_T1-R4: 0.252	MARIMASTAT	CLINICAL TRIALS	DIRECT_TARGET	UNCLASIFIED	0.512	0.174	CANCER	DrugBank
0	15:29716761:T:C	TJP1	BR_LN01: 0.350; SU_T1-R1: 0.663; SU_T1-R2: 0.451; SU_T1-R3: 0.203; SU_T1-R4: 0.263	DEXAMETHASONE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.212	CLINICAL_CANCER	NCI
				RISPERIDONE	APPROVED	BIOMARKER	UNCLASIFIED	0.211	0.212	nan	PharmGKB
				ETHANOL	CLINICAL TRIALS	BIOMARKER	UNCLASIFIED	0.422	0.212	CANCER	NCI
0	16:55692026:G:A	SLC6A2	BR_LN01: 0.299; SU_T1-R1: 0.360; SU_T1-R2: 0.279; SU_T1-R3: 0.130; SU_T1-R4: 0.232	BUPROPION	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.715	0.355	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TEND; TdgClinicalTrial
				IMIPRAMINE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.715	0.355	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TEND; TdgClinicalTrial
				METHYLPHENIDATE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.715	0.355	CLINICAL_CANCER	DrugBank; GuideToPharmacology; PharmGKB; TEND; TdgClinicalTrial
0	16:66400854:G:A	CDH5	BR_LN01: 0.302; SU_T1-R1: 0.340; SU_T1-R2: 0.296; SU_T1-R3: 0.247; SU_T1-R4: 0.243	LENALIDOMIDE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.912	0.326	BLOOD; BONE_MARROW	DrugBank
0	7:40078745:T:C	CDK13	BR_LN01: 0.499; SU_T1-R1: 0.494; SU_T1-R2: 0.436; SU_T1-R3: 0.375; SU_T1-R4: 0.304	PHA-793887	CLINICAL TRIALS	BIOMARKER	UNCLASIFIED	0.412	0.352	CANCER	ChEMBLInteractions
				602306-29-6	CLINICAL TRIALS	BIOMARKER	UNCLASIFIED	0.012	0.352	nan	ChEMBLInteractions
				RONICICLIB	CLINICAL TRIALS	BIOMARKER	UNCLASIFIED	0.012	0.352	nan	ChEMBLInteractions
0	8:104424341:T:A	DPYS	BR_LN01: 0.256; SU_T1-R1: 0.249; SU_T1-R2: 0.338; SU_T1-R3: 0.112; SU_T1-R4: 0.218	DEXRAZOXANINE	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.312	BREAST	NCI

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				5-FLUOROURACIL	APPROVED	BIOMARKER	UNCLASIFIED	0.811	0.312	BREAST; COLON; PANCREAS; RECTUM; SKIN; STOMACH	PharmGKB
				ATENOLOL	APPROVED	BIOMARKER	UNCLASIFIED	0.221	0.312	nan	PharmGKB
0	8:40674897:C:T	ZMAT4	BR_LN01: 0.191; SU_T1-R1: 0.393; SU_T1-R2: 0.376; SU_T1-R3: 0.229; SU_T1-R4: 0.200	ATENOLOL	APPROVED	BIOMARKER	UNCLASIFIED	0.221	0.188	nan	PharmGKB
1	12:51721732:C:T	SCN8A	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.035; SU_T1-R4: 0.000	OXCARBAPINE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.292	CLINICAL_CANCER	ChEMBLInteractions
				RILUZOLE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.292	CLINICAL_CANCER	ChEMBLInteractions
				PHENYTOIN	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.313	0.292	nan	ChEMBLInteractions; DrugBank
1	15:29719125:T:G	TJP1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.004; SU_T1-R3: 0.063; SU_T1-R4: 0.000	DEXAMETHASONE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.212	CLINICAL_CANCER	NCI
				RISPERIDONE	APPROVED	BIOMARKER	UNCLASIFIED	0.211	0.212	nan	PharmGKB
				ETHANOL	CLINICAL_TRIALS	BIOMARKER	UNCLASIFIED	0.422	0.212	CANCER	NCI
1	1:239908667:G:A	CHRM3	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.156; SU_T1-R4: 0.000	PILOCARPINE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.915	0.278	HEAD_AND_NECK	ChEMBLInteractions; DrugBank; GuideToPharmacology; TdgClinicalTrial
				PILOCARPINE HYDROCHLORIDE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.912	0.278	HEAD_AND_NECK	ChEMBLInteractions
				OLANZAPINE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.714	0.278	CLINICAL_CANCER	DrugBank; TEND; TdgClinicalTrial
1	2:159412362:T:G	BAZ2B	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.191; SU_T1-R4: 0.000	TAMOXIFEN	APPROVED	BIOMARKER	UNCLASIFIED	0.822	0.240	BREAST	DTC

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				DOXORUBICIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.812	0.240	ADRENAL_GLAND; BLADDER; BLOOD; BLOOD_AND_LYMPH_VESSELS; BONE; BONE_MARROW; BREAST; KIDNEY; LUNG; OVARY; SOFT_TISSUE; STOMACH; THYROID	DTC
				LIOTHYRONINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.812	0.240	THYROID	DTC
1	2:27523281:G:A	GCKR	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.154; SU_T1-R4: 0.000	ETHANOL	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.421	0.377	CANCER	PharmGKB
1	5:45303776:G:T	HCN1	BR_LN01: 0.000; SU_T1-R1: 0.001; SU_T1-R2: 0.000; SU_T1-R3: 0.040; SU_T1-R4: 0.001	IVABRADINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.292	CLINICAL_CANCER	GuideToPharmacology
				TRAMADOL	APPROVED	BIOMARKER	UNCLASSIFIED	0.221	0.292	nan	PharmGKB
1	7:105101587:C:T	KMT2E	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.005; SU_T1-R3: 0.054; SU_T1-R4: 0.000	DULOXETINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.211	0.296	nan	PharmGKB
2	10:69097231:C:T	SRGN	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	ENOXAPARIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.622	0.256	CLINICAL_CANCER	NCI
2	11:20626756:C:T	SLC6A5	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.011; SU_T1-R4: 0.053	HALOPERIDOL	APPROVED	BIOMARKER	UNCLASSIFIED	0.631	0.325	CLINICAL_CANCER	PharmGKB
				BITOPERTIN	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.012	0.325	nan	GuideToPharmacology
2	11:63904880:G:A	MARK2	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.033	FOSTAMATINIB	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.732	0.245	CLINICAL_CANCER	DrugBank

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				ACETAMINOPHEN	APPROVED	BIOMARKER	UNCLASSIFIED	0.212	0.245	nan	NCI
				CENISERTIB	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.422	0.245	CANCER	DTC
2	12:10447587:G:A	KLRC1	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	MONALIZUMAB	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASSIFIED	0.512	0.387	CANCER	GuideToPharmacology; SAbDab; TTD
2	12:2674604:G:A	CACNA1C	BR_LN01: 0.000; SU_T1-R1: 0.004; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	VERAPAMIL	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.716	0.279	CLINICAL_CANCER	ChEMBLInteractions; DrugBank; GuideToPharmacology; PharmGKB; TEND; TdgClinicalTrial
				FISH OIL	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.279	CLINICAL_CANCER	DrugBank
				ISAVUCONAZOLE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.279	CLINICAL_CANCER	DrugBank
2	12:2679489:G:A	CACNA1C	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.001; SU_T1-R3: 0.000; SU_T1-R4: 0.041	VERAPAMIL	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.716	0.279	CLINICAL_CANCER	ChEMBLInteractions; DrugBank; GuideToPharmacology; PharmGKB; TEND; TdgClinicalTrial
				FISH OIL	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.279	CLINICAL_CANCER	DrugBank
				ISAVUCONAZOLE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.712	0.279	CLINICAL_CANCER	DrugBank
2	17:35001332:C:T	LIG3	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.002; SU_T1-R4: 0.000	BLEOMYCIN	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.912	0.458	BLOOD; HEAD_AND_NECK; OVARY; PENIS; TESTIS; UTERUS; VULVA	DrugBank
				CISPLATIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.821	0.458	BLADDER; OVARY; TESTIS	PharmGKB
				CYCLOPHOSPHAMIDE	APPROVED	BIOMARKER	UNCLASSIFIED	0.811	0.458	ADRENAL_GLAND; BLOOD; BONE_MARROW; BREAST; EYE; OVARY	PharmGKB
2	17:42712363:G:A	EZH1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.004; SU_T1-R3: 0.000; SU_T1-R4: 0.049	TAZEMETOSTAT	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.912	0.546	BLOOD; SOFT_TISSUE	DrugBank
2	17:80107556:G:A	GAA	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.013; SU_T1-R4: 0.128	DOXAZOSIN MESYLATE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.315	CLINICAL_CANCER	DTC

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				OLANZAPINE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.315	CLINICAL_CANCER	DTC
				ACARBOSE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.325	0.315	nan	DTC; DrugBank; TEND; TdgClinicalTrial
2	19:38499812:G:A	RYR1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	CAFFEINE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.723	0.338	CLINICAL_CANCER	DrugBank; GuideToPharmacology; PharmGKB
				HALOPERIDOL	APPROVED	BIOMARKER	UNCLASIFIED	0.631	0.338	CLINICAL_CANCER	PharmGKB
				DESFLURANE	APPROVED	BIOMARKER	UNCLASIFIED	0.611	0.338	CLINICAL_CANCER	PharmGKB
2	20:49524207:G:A	PTGIS	BR_LN01: 0.000; SU_T1-R1: 0.002; SU_T1-R2: 0.000; SU_T1-R3: 0.015; SU_T1-R4: 0.025	EPOPROSTENOL	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.312	0.384	nan	DrugBank
2	2:10440743:G:A	ODC1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	VINCISTINE	APPROVED	BIOMARKER	UNCLASIFIED	0.822	0.351	ADRENAL_GLAND; BLOOD; KIDNEY; SOFT_TISSUE	DTC
				EFLORNITHINE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.714	0.351	CLINICAL_CANCER	DrugBank; GuideToPharmacology; TTD; TdgClinicalTrial
				96020-91-6	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.712	0.351	CLINICAL_CANCER	ChEMBLInteractions
2	2:167246592:A:G	XIRP2	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.046	ATENOLOL	APPROVED	BIOMARKER	UNCLASIFIED	0.221	0.270	nan	PharmGKB
2	2:238150886:G:A	KLHL30	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	APR-246	CLINICAL_TRIALS	BIOMARKER	UNCLASIFIED	-0.411	0.253	CANCER	GDSC
2	3:179221146:G:A	PIK3CA	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.029; SU_T1-R4: 0.055	ALPELISIB	APPROVED	DIRECT_TARGET	ONC	0.920	0.799	BREAST	CancerCommons; ChEMBLInteractions; ClarityFoundationBiomarkers; DrugBank; FDA; GDSC; GuideToPharmacology; JAX-CKB; MOAImanac; MyCancerGenome; MyCancerGenomeClinicalTrial; PharmGKB; TTD

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				COPANLISIB	APPROVED	DIRECT_TARGET	ONC	0.919	0.799	BLOOD	CancerCommons; ChEMBLInteractions; ClarityFoundationBiomarkers; DrugBank; GuideToPharmacology; JAX-CKB; MyCancerGenome; MyCancerGenomeClinicalTrial
				ALPELISIB + FULVESTRANT	APPROVED	DIRECT_TARGET	ONC	0.913	0.799	BREAST	CIViC; OncoKB
2	4:55456299:T:C	CLOCK	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.107	7439-93-2	APPROVED	BIOMARKER	UNCLASSIFIED	0.621	0.260	CLINICAL_CANCER	PharmGKB
2	5:95517182:A:G	SKIC3	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.047; SU_T1-R3: 0.000; SU_T1-R4: 0.000	DULOXETINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.211	0.125	nan	PharmGKB
2	7:142021631:G:A	MGAM	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.001; SU_T1-R3: 0.000; SU_T1-R4: 0.000	MIGLITOL	APPROVED	DIRECT_TARGET	ONC	0.325	0.313	nan	DrugBank; GuideToPharmacology; TEND; TTD; TdgClinicalTrial
				ACARBOSE	APPROVED	DIRECT_TARGET	ONC	0.324	0.313	nan	DrugBank; TEND; TTD; TdgClinicalTrial
				33626-08-3	CLINICAL_TRIALS	BIOMARKER	ONC	0.412	0.313	CANCER	DTC
4	1:22906103:G:A	EPHB2	BR_LN01: 0.171; SU_T1-R1: 0.101; SU_T1-R2: 0.262; SU_T1-R3: 0.074; SU_T1-R4: 0.128	VANDETANIB	APPROVED	BIOMARKER	TSG	0.812	0.339	THYROID	ChEMBLInteractions
				FOSTAMATINIB	APPROVED	DIRECT_TARGET	TSG	0.712	0.339	CLINICAL_CANCER	DrugBank
				MK-8353	CLINICAL_TRIALS	DIRECT_TARGET	TSG	0.511	0.339	CANCER	TTD
4	2:73266617:C:T	FBXO41	BR_LN01: 0.110; SU_T1-R1: 0.201; SU_T1-R2: 0.153; SU_T1-R3: 0.019; SU_T1-R4: 0.056	2-PHENYLETHYL ISOTHIOCYANATE	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASSIFIED	0.512	0.205	CANCER	DrugBank
5	11:14644797:C:T	PDE3B	BR_LN01: 0.208; SU_T1-R1: 0.189; SU_T1-R2: 0.000; SU_T1-R3: 0.049; SU_T1-R4: 0.048	AMINOPHYLLINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.278	CLINICAL_CANCER	ChEMBLInteractions
				PENTOXIFYLLINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.278	CLINICAL_CANCER	ChEMBLInteractions
				THEOPHYLLINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.278	CLINICAL_CANCER	ChEMBLInteractions

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
5	14:47059299:C:T	MDGA2	BR_LN01: 0.110; SU_T1-R1: 0.110; SU_T1-R2: 0.002; SU_T1-R3: 0.030; SU_T1-R4: 0.042	FLUVOXAMINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.611	0.185	CLINICAL_CANCER	PharmGKB
				MILNACIPRAN	APPROVED	BIOMARKER	UNCLASSIFIED	0.211	0.185	nan	PharmGKB
				PAROXETINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.211	0.185	nan	PharmGKB
5	17:82089067:C:T	FASN	BR_LN01: 0.124; SU_T1-R1: 0.124; SU_T1-R2: 0.000; SU_T1-R3: 0.036; SU_T1-R4: 0.049	DOXORUBICIN	APPROVED	BIOMARKER	ONC	0.812	0.273	ADRENAL_GLAND; BLADDER; BLOOD; BLOOD_AND_LYMPH_VESSELS; BONE; BONE_MARROW; BREAST; KIDNEY; LUNG; OVARY; SOFT_TISSUE; STOMACH; THYROID	NCI
				LIOTHYRONINE	APPROVED	BIOMARKER	ONC	0.812	0.273	THYROID	NCI
				CLOZAPINE	APPROVED	BIOMARKER	ONC	0.612	0.273	CLINICAL_CANCER	NCI
5	2:108010631:G:A	SLC5A7	BR_LN01: 0.173; SU_T1-R1: 0.149; SU_T1-R2: 0.000; SU_T1-R3: 0.034; SU_T1-R4: 0.022	(S)-ONDANSETRON	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.177	CLINICAL_CANCER	DTC
				ONDANSETRON HYDROCHLORIDE	APPROVED	BIOMARKER	UNCLASSIFIED	0.612	0.177	CLINICAL_CANCER	DTC
				ACETYLCHOLINE CHLORIDE	APPROVED	BIOMARKER	UNCLASSIFIED	0.212	0.177	nan	DTC
5	2:137616241:A:C	THSD7B	BR_LN01: 0.142; SU_T1-R1: 0.180; SU_T1-R2: 0.000; SU_T1-R3: 0.027; SU_T1-R4: 0.041	ETHANOL	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.411	0.145	CANCER	PharmGKB
5	7:108078280:G:T	LAMB4	BR_LN01: 0.221; SU_T1-R1: 0.187; SU_T1-R2: 0.002; SU_T1-R3: 0.014; SU_T1-R4: 0.035	OCRIPLASMIN	APPROVED	BIOMARKER	UNCLASSIFIED	0.212	0.178	nan	ChEMBLInteractions

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
6	13:113117542:G:A	F7	BR_LN01: 0.027; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	NONACOG BETA PEGOL	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.312	0.401	nan	DrugBank
				RNAPC2	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASIFIED	0.112	0.401	nan	DrugBank
6	18:54293826:G:A	POLI	BR_LN01: 0.034; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.002	CISPLATIN	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.250	BLADDER; OVARY; TESTIS	DTC
				EPIRUBICIN HYDROCHLORIDE	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.250	BREAST	DTC
				IDARUBICIN	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.250	BLOOD	DTC
6	4:154744545:G:C	LRAT	BR_LN01: 0.027; SU_T1-R1: 0.001; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	RETINOL	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.312	0.228	nan	DrugBank
6	6:36684352:G:A	CDKN1A	BR_LN01: 0.034; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	PARAPLATIN	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.344	OVARY	NCI
				5-FLUOROURACIL	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.344	BREAST; COLON; PANCREAS; RECTUM; SKIN; STOMACH	NCI
				FLUTAMIDE	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.344	PROSTATE	NCI
6	7:45648745:G:A	ADCY1	BR_LN01: 0.025; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	7439-93-2	APPROVED	BIOMARKER	UNCLASIFIED	0.621	0.227	CLINICAL_CANCER	PharmGKB
				HYOSCYAMINE	CLINICAL_TRIALS	DIRECT_TARGET	UNCLASIFIED	0.112	0.227	nan	DrugBank
6	8:32760291:A:C	NRG1	BR_LN01: 0.047; SU_T1-R1: 0.000; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	TAMOXIFEN	APPROVED	BIOMARKER	UNCLASIFIED	0.822	0.276	BREAST	NCI
				VINCISTINE	APPROVED	BIOMARKER	UNCLASIFIED	0.812	0.276	ADRENAL_GLAND; BLOOD; KIDNEY; SOFT_TISSUE	NCI

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				CETUXIMAB	APPROVED	BIOMARKER	UNCLASIFIED	0.811	0.276	COLON ; HEAD_AND_NECK; RECTUM	PharmGKB
7	2:135812488:G:A	LCT	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.172; SU_T1-R3: 0.000; SU_T1-R4: 0.000	GLUCONOLACTONE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.312	0.384	nan	DrugBank
7	4:46041252:C:T	GABRG1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.149; SU_T1-R3: 0.000; SU_T1-R4: 0.003	MIDAZOLAM	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.714	0.275	CLINICAL_CANCER	ChEMBLInteractions; TEND; TdgClinicalTrial
				DESFLURANE	APPROVED	BIOMARKER	UNCLASIFIED	0.612	0.275	CLINICAL_CANCER	ChEMBLInteractions
				ALPRAZOLAM	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.314	0.275	nan	ChEMBLInteractions; TEND; TdgClinicalTrial
7	7:100098257:T:A	MCM7	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.189; SU_T1-R3: 0.000; SU_T1-R4: 0.002	SAVOLITINIB	CLINICAL_TRIALS	BIOMARKER	UNCLASIFIED	0.411	0.275	CANCER	GDSC
7	9:134727356:C:T	COL5A1	BR_LN01: 0.000; SU_T1-R1: 0.000; SU_T1-R2: 0.235; SU_T1-R3: 0.000; SU_T1-R4: 0.006	OCRIPLASMIN	APPROVED	BIOMARKER	UNCLASIFIED	0.212	0.221	nan	ChEMBLInteractions
				DACTINOMYCIN	APPROVED	BIOMARKER	UNCLASIFIED	-0.811	0.221	BONE; KIDNEY; SOFT_TISSUE; TESTIS	GDSC
				DACTOLISIB	CLINICAL_TRIALS	BIOMARKER	UNCLASIFIED	-0.421	0.221	CANCER	GDSC
9	12:50074127:C:G	ASIC1	BR_LN01: 0.000; SU_T1-R1: 0.042; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	AMILORIDE	APPROVED	DIRECT_TARGET	UNCLASIFIED	0.313	0.200	nan	DrugBank; GuideToPharmacology; TTD
				IBUPROFEN	APPROVED	BIOMARKER	UNCLASIFIED	0.232	0.200	nan	GuideToPharmacology
				NAFAMOSTAT	CLINICAL_TRIALS	BIOMARKER	UNCLASIFIED	0.022	0.200	nan	GuideToPharmacology
9	16:30121832:C:G	MAPK3	BR_LN01: 0.000; SU_T1-R1: 0.022; SU_T1-R2: 0.002; SU_T1-R3: 0.000; SU_T1-R4: 0.000	ETOPOSIDE	APPROVED	BIOMARKER	ONC	0.822	0.375	LUNG; TESTIS	DTC

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				CYCLOPHOSPHAMIDE	APPROVED	BIOMARKER	ONC	0.812	0.375	ADRENAL_GLAND; BLOOD; BONE_MARROW; BREAST; EYE; OVARY	NCI
				ARSENIC TRIOXIDE	APPROVED	DIRECT_TARGET	ONC	0.712	0.375	CLINICAL_CANCER	DrugBank
9	16:53459467:T:A	RBL2	BR_LN01: 0.000; SU_T1-R1: 0.070; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	RAPAMYCIN	APPROVED	BIOMARKER	TSG	0.612	0.220	CLINICAL_CANCER	NCI
9	17:81906819:C:T	PCYT2	BR_LN01: 0.000; SU_T1-R1: 0.143; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	LAMIVUDINE	APPROVED	BIOMARKER	UNCLASSIFIED	0.211	0.254	nan	PharmGKB
9	18:63318123:T:C	BCL2	BR_LN01: 0.000; SU_T1-R1: 0.029; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	VENETOCLAX	APPROVED	DIRECT_TARGET	ONC	0.925	0.435	BLOOD	ChEMBLInteractions; DrugBank; GuideToPharmacology; MyCancerGenome; TTD
				DOCETAXEL	APPROVED	DIRECT_TARGET	ONC	0.923	0.435	BREAST; ESOPHAGUS; HEAD_AND_NECK; LUNG; PROSTATE; STOMACH	DrugBank; PharmGKB; TdgClinicalTrial
				BORTEZOMIB	APPROVED	DIRECT_TARGET	ONC	0.922	0.435	BLOOD; BONE_MARROW	TALC
9	19:38840188:T:C	HNRNPL	BR_LN01: 0.000; SU_T1-R1: 0.055; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	7440-50-8	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.722	0.165	CLINICAL_CANCER	DrugBank
9	4:155711048:G:C	GUCY1A1	BR_LN01: 0.000; SU_T1-R1: 0.027; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	ISOSORBIDE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.322	0.228	nan	DrugBank
				ISOSORBIDE DINITRATE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.228	nan	ChEMBLInteractions
				ISOSORBIDE MONONITRATE	APPROVED	DIRECT_TARGET	UNCLASSIFIED	0.312	0.228	nan	ChEMBLInteractions
9	7:100892698:C:T	ACHE	BR_LN01: 0.000; SU_T1-R1: 0.131; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	BETAMETHASONE	APPROVED	BIOMARKER	TSG	0.812	0.297	BLOOD	NCI

Clone	Mutation ID	Gene Symbol	VAF	Drug	Status	Interaction Type	Driver Gene	dScore	gScore	Cancer	Source
				THIOGUANIN E	APPROVED	BIOMARKER	TSG	0.812	0.297	BLOOD	NCI
				METHYLENE BLUE CATION	APPROVED	DIRECT_TARGET	TSG	0.712	0.297	CLINICAL_CANCER	DrugBank
9	7:99929382:G:-	GJC3	BR_LN01: 0.000; SU_T1-R1: 0.114; SU_T1-R2: 0.000; SU_T1-R3: 0.000; SU_T1-R4: 0.000	1-OCTANOL	CLINICAL_TRIALS	BIOMARKER	UNCLASSIFIED	0.012	0.377	nan	GuideToPharmacology

Glossary of Terms

Clone

A population of tumor cells that all share a specific set of mutations inherited from a common ancestor cell. Ideally, each inferred mutation cluster would correspond to a biologically distinct clone.

Cluster

Groups of cancer cells that share a common set of somatic mutations, which are inferred by clustering mutations with similar cellular prevalence estimates across one or more tumor samples.

Driver Gene

Genes whose mutations are responsible for the development and progression of cancer.

dScore

Pandrug's indicator which measures the suitability of the treatment for a particular patient. It ranges from -1 to 1, with the negative values corresponding to resistance and the positive values corresponding to sensitivity.

gScore

Pandrug's indicator which measures the biological relevance of a gene in the tumoral process and its druggability. It ranges from 0 to 1, with higher values corresponding to more relevant and actionable targets.

Interaction Type

Drugs approved for cancer treatment were manually classified into 7 different groups: chemotherapy, targeted therapy, hormone therapy, immunotherapy, photodynamic therapy, combination therapy, and other.

Pandrug

A bioinformatics platform to prioritize anticancer drug treatments according to individual multi-omics data.

Parent cells

Group of cells that are set to be the origin of the tumor's phylogeny while performing the clustering and the clonal tree. The estimation of the proportions of parent cells may vary depending on the samples provided.

Study

Group of biologically related samples that are analyzed together in Clonucopy. The experimental design and grouping criteria are defined according to specific research objectives, allowing multiple use cases adapted to the needs of the comparative analysis.

Variant Allele Frequencies (VAF)

A biomarker that expresses the proportion of sequencing reads that support a specific variant allele relative to the total number of reads within a genomic locus. It ranges from 0 to 1.